BROADER ECONOMY IN TRANSITION · BASE SCENARIOS (SSP/RCP)

What are the environmental consequences of micro- and meso-level decisions?
designing, operating, deploying or adopting technologies, in diverse contexts and over the transition

which units operate?

how they operate?

FROM KNOWN TO UNKNOWN

Develop and evaluate against a reference model

Confront with observed and reported data

Extend where no full reference exists

Pillar I · Market consequences
Which suppliers respond, when and where, with what environmental profile

PhD I · Task 1.1
Building a scenario-consistent framework for marginal suppliers, from model perturbation to heuristics

PhD I · Task 1.1
Testing marginal-supplier heuristics against structural model responses

MSc I · Task 1.2
Testing heuristics against past metal capacity, then deriving marginal supply from dynamic MFA

MSc III · Task 1.3
Disaggregating regional scenarios to the local markets where decisions play out

Joint strategy
local markets and processes as instances of broader ones; simplifications tested

Energy
Pillar I case

Metallurgy
shared case

Fertilizer
shared case

Pillar II · Process consequences
How processes and their environmental flows respond to design, operation and context

PhD II · Task 2.1
Wrapping process models of any sophistication in an interoperable interface, instantiated by scenario and decision

PhD III · Task 2.2
AI-assisted model building, evaluated where the answer is known

MSc II · Task 2.3
Calibrating engineering models against reported plant performance, and testing what generic datasets miss

PhD III · Task 2.2
Representing an emerging process before deployment, as alternative configurations

OUTCOMES

Scenario-consistent marginal suppliers
and the errors of common heuristics, characterized

Cross-case synthesis
joint assessment of market and process effects

Interoperable process-model classes
and an evaluated workflow to build more

Consistent, discriminating prospective consequential LCA for environmental decisions across contexts
under which conditions representation choices change the decisions supported, in the cases studied

TIMING
Years of activity, 1 to 5
1 2 3 4 5